Notes: Asker, Collard-Wexler, and De Loecker (JPE, 2014)

Dynamic Inputs and Resource (Mis)Allocation

Contents

1 Big picture

- **Question.** How much of the cross-sectional dispersion in the marginal revenue product of capital (MRPK) should we interpret as true resource misallocation, and how much can be explained by efficient firm behavior when capital is a dynamic input with adjustment costs?
- **Setting.** The paper studies manufacturing producers in eight country-specific data sets (the United States, Chile, France, India, Mexico, Romania, Slovenia, and Spain) and then extends the cross-country analysis to a World Bank Enterprise Survey sample covering 33 developing countries.
- **Core challenge.** Static misallocation measures treat dispersion in MRPK as evidence of distortions. But if firms face capital adjustment costs and receive persistent productivity shocks, a capital stock chosen in the past can look ex post "wrong" even when the dynamic allocation is fully optimal.
- **Main theoretical result.** In a standard dynamic investment model with one-period time to build and capital adjustment costs, higher volatility of productivity generates higher cross-sectional dispersion in the *static* marginal revenue product of capital.
- **Main empirical result.** Industries and countries with more volatile revenue productivity (TFPR) have systematically greater dispersion in MRPK. This pattern holds within countries, across countries, and across alternative volatility measures.
- **Main structural result.** A simple model that varies the estimated AR(1) productivity process across industry-country cells can explain a large share of the observed dispersion in MRPK. Depending on the specification, the model fit is on the order of an uncentered R^2 of about 0.8 to 0.9.
- **Economic takeaway.** Measured "misallocation" in capital may partly reflect efficient dynamic responses to uncertainty, not just policy distortions or financial frictions.
- **Why this paper matters.** It is a benchmark paper for anyone using MRPK dispersion to study development, misallocation, and productivity. The paper shows that dynamics and volatility are not second-order details - they fundamentally change how static wedges should be interpreted.

2 Data and measurement (key objects)

2.1 Tier 1 producer-level panels

- **Unit of observation.** Producer i in year t, where the unit is a plant in the US, Chile, and Mexico data and a firm in France, India, Romania, Slovenia, and Spain.
- **Countries and periods.** The main data sets are US manufacturing plants from 1972–97, Chilean plants from 1979–86, French/Romanian/Spanish firms from 1999–2007, Indian firms from 1989–2003, Mexican plants from 1984–90, and Slovenian firms from 1994–2000.
- **Observed variables.** Sales or output, employment, intermediate inputs, investment, and fixed assets or capital stock. Industries are defined as narrowly as the source data allow (for example, four-digit SIC in the United States and two-digit NACE in several European data sets).

- **Coverage.** The tier 1 data are large manufacturing panels used extensively in the productivity literature. The US sample alone contains 735,342 plants over 26 years after applying data-quality and disclosure filters.

2.2 Tier 2 World Bank Enterprise Survey sample

- **Cross-country extension.** The World Bank Enterprise Survey (WBES) provides a broader cross section of developing countries.
- **Usable sample.** Out of 41 surveyed countries, 33 have enough information on capital, sales, materials, and labor over multiple years to compute TFPR changes and capital changes.
- **Final sample size.** The usable WBES sample contains 5,558 firms. These firms are not representative of the full economy and tend to be relatively large, but the data are useful for cross-country patterns in volatility and MRPK dispersion.

2.3 Measurement of TFPR and MRPK

The paper starts from a constant-returns production function and a constant-elasticity demand curve:

$$Q_{it} = A_{it} K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\alpha_M}, \qquad Q_{it} = B_{it} P_{it}^{-\varepsilon}.$$

Combining these two equations gives a sales-generating production function:

$$S_{it} = Q_{it} K_{it}^{\beta_K} L_{it}^{\beta_L} M_{it}^{\beta_M},$$

where

$$Q_{it} = A_{it}^{1-1/\varepsilon} B_{it}^{1/\varepsilon}, \qquad \beta_X = \alpha_X \left(1 - \frac{1}{\varepsilon}\right) \quad \text{for } X \in \{K, L, M\}.$$

The paper defines revenue productivity (TFPR) as

$$q_{it} = \log(Q_{it}).$$

The static marginal revenue product of capital is

$$MRPK_{it} = \log(\beta_K) + \log(S_{it}) - \log(K_{it}) = \log(\beta_K) + s_{it} - k_{it}.$$

- **Interpretation.** The paper works with *revenue*-based productivity, not physical productivity. Hence measured TFPR can reflect markups, demand shifts, policy distortions, or adjustment costs.
- **Cost-share measurement.** For labor and materials, the revenue production coefficients are measured as median expenditure shares within each industry-country cell:

$$\beta_X^{sc} = \text{median}\left\{\frac{P_{it}^X X_{it}}{S_{it}} : i \in sc\right\}, \qquad X \in \{L, M\}.$$

- **Capital coefficient.** Given constant returns in the physical production function, the revenue coefficients sum to $(\varepsilon - 1)/\varepsilon$, so

$$\beta_K = \frac{\varepsilon - 1}{\varepsilon} - \beta_L - \beta_M.$$

  The baseline calibration sets $\varepsilon = 4$.

2.4 Unobservables, timing, and moments of interest

- **Productivity process.** TFPR follows an AR(1):

$$q_{it} = \mu + \rho q_{i,t-1} + \sigma n_{it}, \qquad n_{it} \sim N(0,1).$$

- **Timing.** Labor and materials are flexible inputs chosen within the period; capital is dynamic and predetermined because investment takes time and is costly to adjust.
- **Main moments.** The paper focuses on three moments at the industry-country-year level:

$$\mathrm{SD}_{st}(MRPK_{it}), \qquad \mathrm{SD}_{st}(MRPK_{it} - MRPK_{i,t-1}), \qquad \mathrm{SD}_{st}(k_{it} - k_{i,t-1}).$$

- **Volatility measure.** The baseline reduced-form measure of volatility is

$$\mathrm{SD}_{st}(q_{it} - q_{i,t-1}),$$

  though the paper also considers AR(1)-based volatility measures.
- **Useful theorem.** The moments above are invariant to firm-specific fixed effects in the mean of the productivity process. A different AR(1) constant shifts levels of inputs but not differences such as MRPK dispersion or capital-change dispersion.

2.5 Empirical applications

- **Reduced form within countries.** Regress MRPK dispersion on TFPR volatility across industries and years.
- **Producer-level implication.** Test whether a firm's MRPK moves with unexpected TFPR shocks because capital cannot adjust instantly.
- **Structural analysis.** Estimate adjustment costs and ask whether the model can quantitatively match the observed dispersion in MRPK.
- **Cross-country extension.** Replicate the volatility-dispersion link in WBES data and compare country-level model predictions to country-level data.

3 Models and empirical specifications (all equations + interpretation)

3.1 Technology, demand, and the static misallocation measure

The paper begins with the production technology and demand system

$$Q_{it} = A_{it} K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\alpha_M}, \tag{1}$$

$$Q_{it} = B_{it} P_{it}^{-\varepsilon}, \tag{2}$$

which imply the sales-generating production function

$$S_{it} = Q_{it} K_{it}^{\beta_K} L_{it}^{\beta_L} M_{it}^{\beta_M}.$$

The static marginal revenue product of capital is

$$MRPK_{it} = \log(\beta_K) + s_{it} - k_{it}.$$

- **Interpretation.** In a static frictionless model, profit maximization equalizes the marginal revenue product of each input to its unit cost. Hence dispersion in MRPK is usually read as a sign of distortions.

- **Paper's twist.** That static logic ignores the fact that capital is chosen dynamically. Once capital is predetermined, a firm hit by a productivity shock can temporarily have an MRPK far from the static optimum even when it behaved optimally ex ante.

3.2 Dynamic investment model

Labor and materials can be chosen frictionlessly each period, so conditional on (Q_{it}, K_{it}) the firm chooses them statically. This yields a reduced-form period profit function

$$p(Q_{it}, K_{it}) = \lambda Q_{it}^{1/(\beta_K+\varepsilon^{-1})} K_{it}^{\beta_K/(\beta_K+\varepsilon^{-1})}.$$

Capital evolves according to

$$K_{i,t+1} = (1-\delta)K_{it} + I_{it},$$

where investment is costly because there is a one-period time to build and an adjustment cost function

$$C(I_{it}, K_{it}, Q_{it}) = I_{it} + C_K^F \mathbf{1}\{I_{it} \neq 0\} p(Q_{it}, K_{it}) + C_K^Q K_{it} \left(\frac{I_{it}}{K_{it}}\right)^2.$$

The value function is

$$V(Q_{it}, K_{it}) = \max_{I_{it}} \left\{ p(Q_{it}, K_{it}) - C(I_{it}, K_{it}, Q_{it}) + \beta \mathbb{E}\big[V(Q_{i,t+1}, (1-\delta)K_{it} + I_{it}) \mid Q_{it}\big] \right\}.$$

The productivity process is

$$q_{it} = \mu + \rho q_{i,t-1} + \sigma n_{it}, \qquad n_{it} \sim N(0,1),$$

so the cross-sectional standard deviation of TFPR is

$$\mathrm{SD}(q_{it}) = \frac{\sigma}{\sqrt{1-\rho^2}}.$$

- **Economic meaning of the adjustment costs.** C_K^F captures lumpy, fixed disruption costs of adjusting capital, while C_K^Q captures smooth convex costs that rise with the investment rate.

- **Why one-period time to build matters.** Investment chosen today affects next period's capital stock. So a productivity shock that arrives after the capital choice can mechanically create temporary MRPK dispersion.

- **Why TFPR volatility matters.** More volatile productivity means firms are more likely to find themselves with a capital stock that is no longer statically optimal.

3.3 Main comparative statics from the model

The central theoretical prediction is that higher productivity volatility generates higher dispersion in static MRPK.

For the knife-edge case $\rho = 0$, the paper shows

$$\mathrm{SD}_{st}(MRPK) = \frac{1}{\beta_K + \varepsilon^{-1}}\sigma.$$

For $\rho \neq 0$, the paper studies the model by simulation and finds the same qualitative result: as σ rises, so does $\mathrm{SD}_{st}(MRPK)$.

- **Interpretation.** Volatility increases the wedge between ex post static optimal capital and the actual capital installed earlier.
- **Why the relation is nonlinear.** At first, higher volatility induces more frequent adjustment. But when volatility becomes very high, current productivity becomes a weaker signal of future profitability, so firms adjust less aggressively. This creates the slope changes seen in the simulation figures.
- **Dynamic-efficiency point.** Static dispersion in MRPK can arise naturally even in an undistorted dynamic economy.

3.4 Reduced-form empirical specifications

The main reduced-form specification relates MRPK dispersion to TFPR volatility at the industry-year level:

$$\mathrm{SD}_{st}(MRPK) = a + b\,\mathrm{Vol}_{st} + \text{fixed effects} + u_{st}.$$

The baseline volatility measure is

$$\mathrm{Vol}_{st} = \mathrm{SD}_{st}(q_{it} - q_{i,t-1}).$$

The paper also estimates country-specific specifications with alternative volatility measures based on AR(1) processes.

To test the producer-level mechanism, the paper estimates

$$MRPK_{it} = \gamma_0 + \gamma_1 y_{it} + \gamma_2 k_{it} + \gamma_3 q_{i,t-1} + \gamma_t + \gamma_s + u_{it},$$

where

$$y_{it} = q_{it} - \mathbb{E}(q_{it} \mid \mathcal{I}_{i,t-1})$$

is the unexpected TFPR shock between $t-1$ and t.

For aggregate implications, the paper also regresses

$$\mathrm{SD}_{st}(\Delta MRPK) \qquad \text{and} \qquad \mathrm{SD}_{st}(\Delta k)$$

on volatility.

- **Interpretation of γ_1.** If capital is predetermined, firms that receive bigger positive TFPR shocks should have higher MRPK, even after conditioning on lagged TFPR and capital. The theory predicts $\gamma_1 > 0$.
- **Why these regressions matter.** The main dispersion result could, in principle, be a correlation driven by data quirks. The shock regression and the aggregate-moment regressions pin down the dynamic mechanism more directly.

3.5 Estimating the TFPR process and adjustment costs

For the structural exercise, the paper estimates the industry-country-specific productivity process

$$q_{it} = \mu_{sc} + \rho_{sc} q_{i,t-1} + \sigma_{sc} n_{it}$$

by maximum likelihood.

It then estimates the capital adjustment cost parameters

$$\nu = \{C_K^F, C_K^Q\}$$

by minimum distance. The three moments matched in each country are:

- the share of producers with less than a 5% year-on-year change in capital,
- the share with more than a 20% year-on-year change in capital,
- the standard deviation of year-on-year log capital growth.

Let $\widehat{W}$ denote the moments in the data and $W(\nu)$ the moments predicted by the model. The criterion function is

$$Q(\nu) = \left(\widehat{W} - W(\nu)\right)' W \left(\widehat{W} - W(\nu)\right),$$

with identity weighting matrix in the baseline implementation.

- **Simulation design.** The firm makes monthly decisions in the model, but the simulated moments are aggregated to the annual frequency of the data.
- **Why this is clever.** The adjustment-cost parameters are pinned down by the *dynamics of capital*, not by MRPK directly. The paper then asks whether those independently estimated dynamics can explain MRPK dispersion.

3.6 Model fit metric and structural specifications

To evaluate the model, the paper compares observed MRPK dispersion x to the model prediction $\hat{x}$ using

$$S^2 = 1 - \frac{(x - \hat{x})'(x - \hat{x})}{x'x}.$$

- **Interpretation.** S^2 is an uncentered-R^2-type measure of fit. It is the share of the observed variation captured by the model, although it can be negative if the model overpredicts dispersion badly.
- **Baseline structural experiment.** Hold production coefficients and adjustment costs fixed at US values and allow only the AR(1) TFPR process to vary across industry-country cells.
- **Richer structural experiments.** Allow production coefficients to vary by industry-country, allow country-specific adjustment costs, double US adjustment costs for all countries, or remove adjustment costs except for one-period time to build.

4 Identification and instruments (what makes the estimates credible)

4.1 What the static approach gets wrong

- **Static benchmark.** In a frictionless static model, dispersion in MRPK is evidence that capital is misallocated across firms.
- **Problem.** Capital is not a frictionless static input. It is chosen dynamically and cannot be costlessly reoptimized after each productivity shock.
- **Implication.** A high-MRPK firm is not necessarily constrained or distorted. It may simply be a firm that was hit by a positive productivity shock after capital was installed.
- **Main message.** Dispersion in static marginal products is not itself sufficient evidence of inefficiency.

4.2 What identifies the paper's mechanism

- **Within-country cross-industry variation.** The reduced-form identification comes from comparing industries with different TFPR volatility within the same country and over time.
- **Cross-country variation.** The WBES data add a second source of evidence: countries with more volatile TFPR also have greater MRPK dispersion.
- **Dynamic-input mechanism.** The shock regression links unexpected TFPR changes directly to MRPK, which is the producer-level implication of the model.
- **Independent estimation of dynamics.** Adjustment costs are estimated from capital-change moments, while the TFPR process is estimated from productivity dynamics. The model's success in matching MRPK dispersion is therefore not mechanical.
- **Fixed-effect robustness.** Theorem 1 shows that the key dispersion moments are invariant to firm-specific fixed effects in the mean of TFPR, so the main results are not artifacts of permanent productivity differences.
- **Capital as the key dynamic input.** The paper checks that dispersion in the marginal revenue product of capital is consistently higher than for labor or materials, which supports the modeling choice that capital adjustment costs are first order.

4.3 Relation to the literature

- **Relative to misallocation papers.** Papers such as Hsieh and Klenow read MRPK dispersion as a sign of wedges. This paper shows that a substantial portion of such dispersion can arise endogenously from efficient dynamics.
- **Relative to investment-under-uncertainty models.** The model builds on Dixit-Pindyck, Caballero-Pindyck, Cooper-Haltiwanger, and Bloom by taking the same dynamic investment framework to the data on MRPK dispersion.
- **Conceptual contribution.** The paper does not deny that distortions matter. Instead, it argues that any serious misallocation exercise must first net out the dispersion that is the natural benchmark in a dynamic world with volatile productivity.

5 Tables and figures

5.1 Figure 1: model simulation of MRPK dispersion and volatility

Read:

- The model predicts a clear positive relation between TFPR volatility σ and the dispersion in static MRPK.
- The figure plots this relation for three persistence values, $\rho = 0.65$, 0.85, and 0.94. In all cases, more volatility means more dispersion.
- The slope is not perfectly linear. At moderate volatility, firms adjust capital more often; at high volatility, current TFPR becomes a weaker predictor of future profitability, so the response of capital flattens out.
- This is the paper's central theoretical insight in one picture: what looks like static capital misallocation can be the efficient consequence of dynamic capital adjustment under uncertainty.

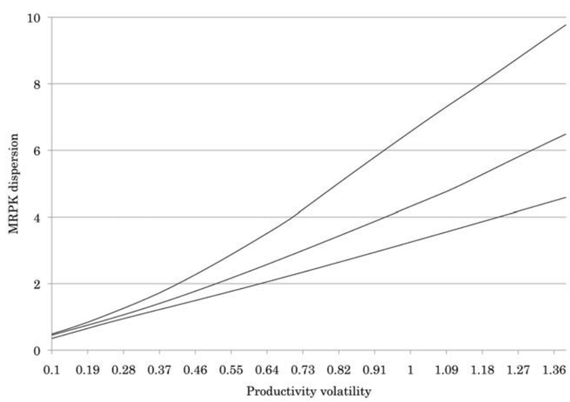

FIG. 1.—MRPK dispersion and volatility: model simulation. Values used in this simulation are $\epsilon = -4$, $\delta = 10$ percent, $\beta = 1/(1 + 0.065)$, $\beta_K = 0.12$, $\beta_M = 0.40$, $\beta_L = 0.23$, $C_K^F = 0.09$, $C_K^Q = 8.8$, $\lambda = 1$, $\mu = 0$, $\rho \in \{0.65, 0.85, 0.94\}$ (corresponding to the lines from bottom [0.65] to top [0.94]), and $\sigma \in [0.1, 1.4]$. We use the means in the US census data to get our β's and use estimates of adjustment costs for the United States discussed in Section V.

TABLE 2
SUMMARY STATISTICS ACROSS TIER 1 DATA SETS

	A. Medians			B. Standard Deviations			
Country	Workers	Δs	$\Delta\omega$	Dispersion MRPK	Dispersion k	Dispersion ω	Volatility
United States[a]	111	.01	.00	.98	1.78	.63	.35
Chile	19	.02	.00	1.22	1.92	.54	.29
France	8	.02	.02	1.28	2.04	.61	.19
India	NA	.06	.04	1.13	1.61	.67	.29
Mexico	141	.02	.02	1.40	2.13	.86	.39
Romania	5	.01	.01	1.38	2.05	.70	.39
Slovenia	4	.07	.03	1.56	2.51	.59	.40
Spain	8	.03	.01	1.48	2.00	.46	.23

NOTE.—Dispersion MRPK is given by SD($MRPK_{it}$), and volatility is SD($\omega_{it} - \omega_{it-1}$); i.e., we compute dispersion across the entire data set.

[a] The median is computed for the US census data as the average of plants between the 48th and 52nd percentiles.

5.2 Table 2: summary statistics across tier 1 data sets

Read:

- The tier 1 data exhibit large cross-country differences in both MRPK dispersion and volatility. MRPK dispersion ranges from 0.98 in the United States to 1.56 in Slovenia.
- Volatility, measured by SD($q_{it} - q_{i,t-1}$), ranges from 0.19 in France to 0.40 in Slovenia.
- The median producer size differs dramatically across data sets because of different collection protocols; for example, the median US plant has 111 workers, while the median French or Spanish firm has only eight.
- These raw moments already suggest that a one-size-fits-all static interpretation of MRPK dispersion is unlikely to be correct.

5.3 Figure 2: volatility and MRPK dispersion in the US data

Read:

- Each point is a four-digit US manufacturing industry. The regression slope is 0.73 with a standard error of 0.08, and the R^2 is about 0.30.
- The positive slope is the simplest reduced-form version of the theory: more volatile industries have greater dispersion in MRPK.
- The United States is the cleanest data environment in the paper, so this figure is an important benchmark before moving to the broader international evidence.

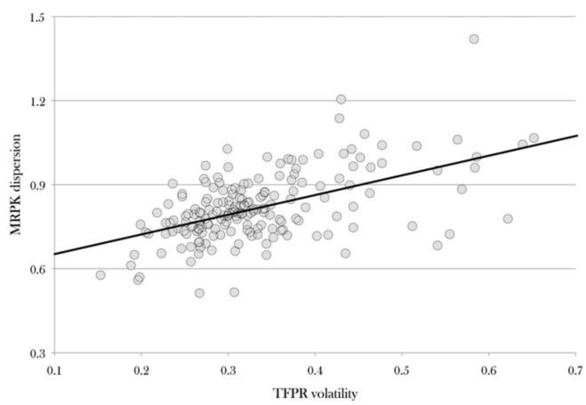

FIG. 2.—Volatility and the dispersion in MRPK: US plant data, 1972–97. The unit of observation is the industry. The line is generated by an OLS regression on 188 industries, in which the estimated slope is 0.73 (0.08) and the constant is 0.57 (0.03), and the R^2 = .3, where the standard errors are in parentheses.

TABLE 3
DISPERSION MRPK AND VOLATILITY

Country	Coefficient	R^2	Industry-Year Observations
United States:			
Plants	.76***	.47	4,037
	(.04)		
Firms	.68***	.44	4,037
	(.07)		
Chile	.54*	.13	55
	(.29)		
France	1.03***	.28	167
	(.33)		
India	.61**	.28	279
	(.17)		
Mexico	.19**	.07	296
	(.07)		
Romania	.44***	.21	126
	(.13)		
Slovenia	.53**	.09	108
	(.21)		
Spain	.56*	.35	181
	(.33)		
All:			
Unweighted	.55***	.67	5,326
	(.15)		
Weighted	.74***	.50	5,326
	(.03)		

NOTE.—We report the coefficient of a regression of SD_{st}(MRPK) against volatility, defined as $SD_{st}(\omega_{it} - \omega_{it-1})$, including year dummies. Standard errors are clustered at the industry level. For all countries, weighted regression includes as weights the number of producers in a country-industry-year observation. These cross-country-industry-year regressions include year and country dummies and report standard errors (in parentheses) clustered at the country level. Table D.15 in the online appendix reports the regression coefficients for the United States using only variation across industries. This is directly related to fig. 2.
* Significant at the 10 percent level.
** Significant at the 5 percent level.
*** Significant at the 1 percent level.

5.4 Table 3: dispersion in MRPK and volatility

Read:

- In every tier 1 country, the coefficient of MRPK dispersion on volatility is positive. For the US data, the coefficient is 0.76 at the plant level and 0.68 at the firm level.
- The relation is statistically significant in every country-specific regression, with coefficients such as 1.03 for France, 0.61 for India, and 0.44 for Romania.
- When all country-industry-year observations are pooled, the coefficient is 0.55 unweighted and 0.74 weighted by the number of producers.

- This table is the main reduced-form result of the paper.

5.5 Table 4: robustness to alternative volatility measures

Read:

- The paper replaces the baseline volatility measure with two alternatives: the innovation variance from an AR(1) process and an AR(1) process with producer fixed effects.
- In 22 of the 24 country-volatility combinations, the coefficient on volatility remains positive and statistically significant at the 10% level or better.
- The different volatility measures are highly correlated - the pairwise correlation is always above 0.72 and often above 0.90.
- So the basic volatility-dispersion result is not an artifact of one particular way of measuring volatility.

TABLE 4
DISPERSION OF MRPK AND VOLATILITY OF TFPR: ROBUSTNESS

	Volatility Measure		
Country	$SD_s(\omega_{it} - \omega_{it-1})$	AR(1)	AR(1)FE
United States	.82***	.86***	1.24***
	(.04)	(.07)	(.11)
Chile	1.48*	2.10***	.33
	(.65)	(.65)	(1.48)
France	1.73***	1.75***	2.55***
	(.41)	(.41)	(.61)
India	1.31***	1.75***	2.75***
	(.33)	(.39)	(.55)
Mexico	.39*	.41**	.33
	(.17)	(.17)	(.25)
Romania	.76***	.94***	1.38*
	(.23)	(.36)	(.72)
Slovenia	2.73***	2.47***	3.47***
	(.41)	(.41)	(.69)
Spain	1.24***	1.46***	2.55***
	(.34)	(.44)	(.59)

NOTE.—We report the coefficient of a regression of SD_{st}(MRPK) against alternative measures of volatility, defined in the text. Standard errors (in parentheses) are clustered at the industry level.
* Significant at the 10 percent level.
** Significant at the 5 percent level.
*** Significant at the 1 percent level.

TABLE 5
ADDITIONAL PREDICTIONS: MRPK AGAINST SHOCKS TO TFPR

Country	Shock	Shock AR(1)	Shock AR(1)FE
United States	1.29***	1.26***	1.13***
	(.00)	(.00)	(.01)
Chile	1.42***	1.42***	1.04***
	(.02)	(.02)	(.02)
France	1.37***	1.37***	1.26***
	(.01)	(.01)	(.01)
India	1.32***	1.32***	1.16***
	(.04)	(.04)	(.04)
Mexico	1.07***	1.07***	.67***
	(.04)	(.04)	(.05)
Romania	1.31***	1.31***	1.12***
	(.01)	(.01)	(.01)
Slovenia	1.65***	1.64***	1.48***
	(.04)	(.04)	(.04)
Spain	1.28***	1.28***	.69***
	(.01)	(.01)	(.01)

NOTE.—We run, by country, ln(MRPK) against the TFPR shock, capital, lagged TFPR, and year and industry fixed effects. The TFPR shock is given by $\xi_{it} \equiv \omega_{it} - E(\omega_{it}|\mathcal{I}_{it-1})$, where $\mathcal{I}_{it-1}$ is the information set of producer i at time $t-1$; depending on the TFPR process, we consider that this contains lagged TFPR and producer and year fixed effects. We suppress the coefficients on capital and lagged TFPR (they are significant with negative and positive signs, respectively, everywhere) and also suppress the fixed effects on year and industry. The standard errors (in parentheses) are clustered at the firm/plant level to account for serially correlation and heteroskedasticity.
*** Significant at the 1 percent level.

5.6 Table 5: MRPK against shocks to TFPR

Read:

- The coefficient on the TFPR shock is positive and significant in every country and every specification. The estimates range from about 1.07 in Mexico to 1.65 in Slovenia in the baseline shock specification.
- This is exactly the producer-level implication of the model: once capital is installed, firms hit by a positive productivity shock should temporarily exhibit higher MRPK.
- The paper also finds that MRPK is mean reverting, with AR(1) coefficients ranging from 0.73 in Romania to 0.90 in Chile.

- Together, these results support the idea that MRPK dispersion is a dynamic object, not a purely static one.

5.7 Figure 3 and Table 6: aggregate implications for $\Delta MRPK$ and Δk

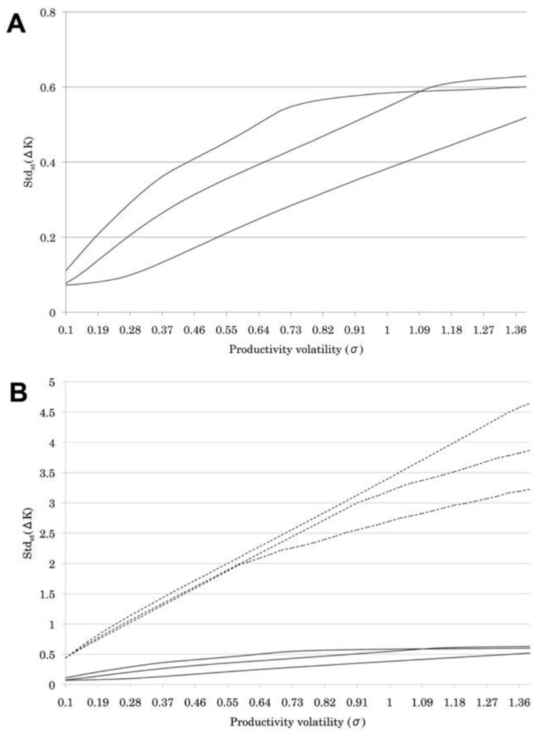

Fig. 3.—Model simulation: dispersion in the change in capital and volatility. Parameters are as for figure 1: $\rho \in \{0.65, 0.85, 0.94\}$ (corresponding to the lines from bottom [0.65] to top [0.94] when $\sigma = 0.3$).

TABLE 6
Aggregate Implications

Aggregate Moment	Coefficient	R^2	Observations
SD_{st}(ΔMRPK):			
United States only	1.03***	.63	4,039
	(.03)		
Excluding the United States	.56***	.64	1,289
	(.07)		
All countries	.89***	.68	5,326
	(.04)		
$\mathrm{SD}_{st}(\Delta k)$:			
United States only	.13***	.31	4,037
	(.02)		
Excluding the United States	.07***	.62	1,182
	(.02)		
All countries	.12***	.76	5,219
	(.02)		
SD$st(\Delta k)$:			
United States only	.17***	.31	4,037
	(.03)		
× {> median vol.}	−.03**		
	(.01)		
Excluding the United States	.19**	.63	1,182
	(.09)		
× {> median vol.}	−.09		
	(.06)		
All countries	.17***	.76	5,219
	(.03)		
× {> median vol.}	−.04**		
	(.02)		

Note.—The coefficients are obtained by regressing each aggregate moment against volatility using country-industry-year variation, where we include year and country fixed effects. Standard errors (in parentheses) are clustered by country when pooled and by industry when using US data.
** Significant at the 5 percent level.
*** Significant at the 1 percent level.

Read:

- The model predicts that volatility should be positively related to the dispersion in changes in MRPK and also to the dispersion in changes in capital.
- Table 6 confirms this. The coefficient of volatility on SD($\Delta MRPK$) is 1.03 for the United States, 0.56 for the pooled non-US sample, and 0.89 for all countries together.
- For capital changes, the coefficients are smaller but still positive: 0.13 for the United States, 0.07 excluding the United States, and 0.12 in the pooled sample.
- Figure 3 shows why the capital-change relation is nonlinear. At higher volatility, firms eventually adjust capital less aggressively because current shocks are less informative about future profitability. The negative interaction term above the median volatility in table 6 is the reduced-form counterpart of that flattening.

5.8 Table 7: comparing dispersion in marginal revenue products across inputs

Read:

- In every country, dispersion in the marginal revenue product of capital is larger than the corresponding dispersion for labor or materials.
- The ranking is the economically natural one: SD($MRPK$) > SD($MRPL$) > SD($MRPM$).
- For example, in the United States the average dispersions are 0.81 for capital, 0.63 for labor, and 0.54 for materials; in Spain they are 1.45, 0.93, and 0.70.
- This table supports the paper's maintained assumption that capital is the input for which adjustment frictions are quantitatively most important.

TABLE 7
COMPARING DISPERSION OF MRPK TO OTHER INPUTS' MARGINAL REVENUE PRODUCTS

	Input		
Country	Capital	Labor	Materials
United States	.81	.63	.54
Chile	1.22	.93	.48
France	1.25	.79	.87
India	1.01	.87	.55
Mexico	1.19	.85	.51
Romania	1.40	1.17	.67
Slovenia	1.54	.98	.54
Spain	1.45	.93	.70

NOTE.—We compute the standard deviation of the marginal revenue product of each input by industry-year, and we report the average across industry-years by country.

TABLE 8
ADJUSTMENT COST ESTIMATES AND MOMENTS BY COUNTRY

	Adjustment Costs		Data Moments on Change in Log Capital		
Country	Convex	Fixed	Less than 5%	More than 20%	Standard Deviation
United States	8.80	.09	.39	.09	.21
Chile	4.10	.07	.19	.11	.28
India	3.46	.12	.29	.19	.30
France	.21	.00	.13	.57	.57
Spain	.74	.00	.20	.41	.59
Mexico	1.15	.22	.08	.73	.66
Romania	.66	.03	.08	.61	.72
Slovenia	.35	.00	.15	.52	.76

NOTE.—Standard errors were computed using the usual formula for minimum-distance estimators. However, because of the large size of the data sets we employ, the standard errors are of the order of 1×10^{-3} or smaller, and so we do not report them. Adjustment costs for Slovenia are based on a model with production function coefficients set to the mean US coefficients (see the discussion in Sec. V.B).

5.9 Table 8: adjustment cost estimates and capital-change moments

Read:

- Adjustment costs differ a lot across countries. The US estimates are a fixed cost of 0.09 and a convex cost of 8.80, while the convex cost is much smaller in Mexico (1.15), Romania (0.66), and Slovenia (0.35).
- The capital-change moments also vary sharply across countries. In the United States, 39% of plants change capital by less than 5% per year, whereas in Romania only 8% do.
- Some countries are estimated to have essentially zero fixed costs beyond the one-period time to build, but the absence of convex costs is strongly rejected by the data.
- For the United States, the fixed cost is economically meaningful - about 1.5 months of output - and the convex cost is large enough that doubling capital within a month would be extremely expensive.

5.10 Table 9: structural fit of the model

Read:

TABLE 9
Dispersion in MRPK, S^2 Measures of Model Fit by Specification

Country	Specification (1)	(2)	(3)	(4)	(5)
United States	.223	.806	.806	.643	.820
France	.892	.702	.899	.944	.651
Chile	.994	.983	.987	.963	.785
India	.984	.941	.964	.976	.596
Mexico	.879	.813	.883	.908	.634
Romania	.983	.923	.817	.702	.846
Slovenia	.967	.774	.967	.984	.683
Spain	.718	.627	.600	.530	.495
All (excluding United States)	.879	.777	.820	.800	.640
All	.674	.786	.816	.748	.696
Specification details:					
All US adjusted costs	X		X		
Own-country adjusted costs		X			
All 2 × US adjusted costs				X	
1-period time to build only					X
US average β's	X				
Industry-country β's		X	X	X	X

Note.—The unit of observation is the country-industry. Specifications are as follows: (1) All countries have the United States' estimated adjustment costs and production coefficients equal to the US averages across industries; (2) industry-country-specific production coefficients (except for Slovenia; see Sec. III.B), country specific adjustment costs, and industry-country-specific AR(1); (3) same as for 2, but with the United States' estimated adjustment costs for all countries; (4) same as for 3, but with twice the United States' estimated adjustment costs for all countries; and (5) same as for 3, but with zero adjustment costs (other than the one-period time to build) for all countries. In all specifications, the AR(1) is estimated using TFPR computed using the production coefficients used in the model specification.

- In the baseline specification that uses US average production coefficients and US adjustment costs for every country, the fit is already strong: $S^2 = 0.674$ pooling all country-industry cells and 0.879 excluding the United States.
- The baseline fit is weakest for the United States itself ($S^2 = 0.223$) because US industries are defined much more narrowly. Allowing industry-country-specific production coefficients raises the US fit to 0.806.
- Across richer specifications, the fit typically lies between about 0.75 and 0.82 when all countries are pooled.
- An important lesson is that *having some adjustment friction matters more than the exact level of the friction*. Doubling US adjustment costs changes fit little, while removing all adjustment costs except time to build lowers the fit.

5.11 Figures 4 and 5: cross-country evidence from the WBES data

Read:

- Figure 4 extends the reduced-form evidence across countries. In the tier 1 country sample, the slope of MRPK dispersion on volatility is about 1.02 with $R^2 = 0.28$. In the WBES sample of 33 countries, the slope is about 0.67 with $R^2 = 0.31$.
- Figure 5 compares country-level MRPK dispersion in the data to the model's prediction. The model fits well: $S^2 = 0.802$ for the WBES countries and 0.906 for the tier 1 countries aggregated to the same level.
- The model tends, if anything, to overpredict MRPK dispersion slightly, which reinforces the paper's broader message that observed dispersion is not prima facie evidence of severe distortions.

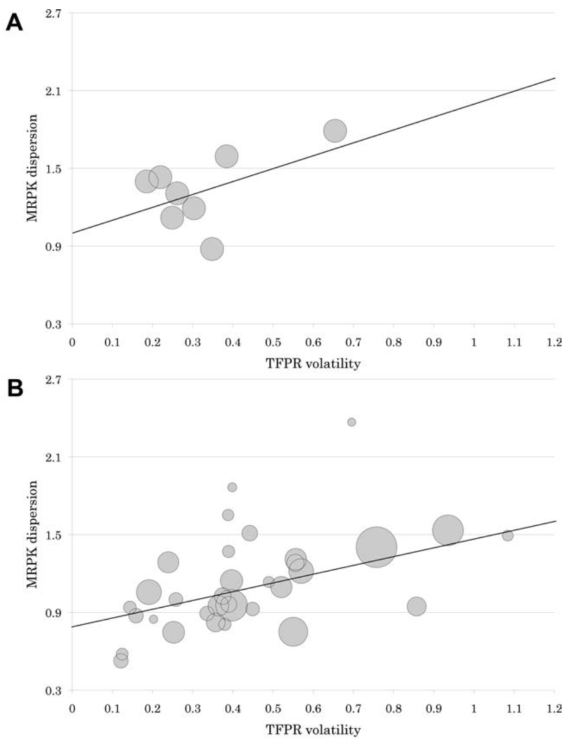

FIG. 4.—Country-level static misallocation and TFPR volatility. Circles indicate countries, where circle size for tier 2 data (panel *B*) is increasing in the number of firms per country. The bold straight line is the line of best fit (computed using OLS with a constant term). The horizontal axis indicates the value of the standard deviation of $\omega_{it} - \omega_{it-1}$. The vertical axis indicates the standard deviation in MRPK. The regression line for panel *A* is given by 1.01 (0.23) + 1.02 (0.66) × vol with an R^2 of .28. The regression line for panel *B* is given by 0.78 (0.10) + 0.67 (0.21) × vol with an R^2 of .31, where standard errors are given in parentheses, and vol denotes our measure of volatility.

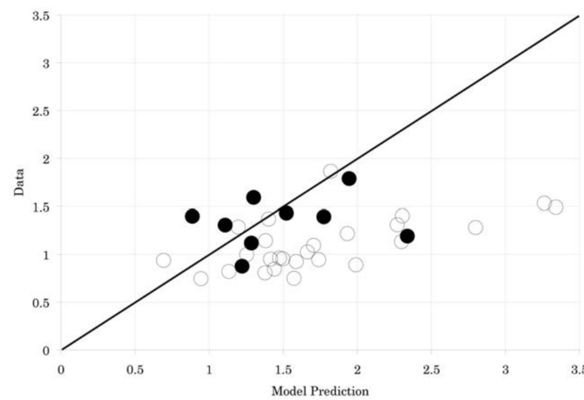

FIG. 5.—Country-level MRPK dispersion: data versus model simulation. The vertical axis is data, while the horizontal axis is the model prediction. The unit of observation is the country. Predictions are computed at the industry-country level and then aggregated to the country level. Black dots are tier 1 countries. Unfilled circles are countries in the WBES data. All predictions use industry-country-specific production coefficients, a country-level AR(1) process, and the adjustment costs estimated for the United States in Section V. The S^2 for the World Bank countries is 0.802 and is 0.906 for the tier 1 countries. The solid line is the 45-degree line.

5.12 Robustness and extensions

Read:

- The main volatility-dispersion pattern is robust to alternative volatility measures, plant versus firm aggregation in the US data, and a wide range of model-specification checks.
- When the authors relate cross-country volatility to external measures, they find that higher costs of contract enforcement are associated with higher TFPR volatility. Natural disasters also have some explanatory power, while political stability has the expected sign but is imprecisely estimated.
- In the WBES cross section, these external measures explain only part of the variation in volatility, which the paper interprets as suggestive rather than definitive.

6 Short summary

- **Conclusion.** Dynamic capital adjustment and productivity volatility can generate substantial dispersion in the static marginal revenue product of capital.
- **Identification insight.** The paper combines reduced-form evidence with a structural dynamic investment model in which the TFPR process and capital adjustment costs are estimated independently from the MRPK dispersion they are later asked to explain.
- **Empirical lesson.** A large share of the observed cross-industry and cross-country variation in MRPK dispersion can be explained without invoking additional distortions. Volatility and capital dynamics are first-order determinants of measured misallocation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Dispersion in static measures of capital misallocation is strongly shaped by whether capital is a dynamic input and by how volatile productivity is.
- **Methodological contribution.** The paper provides a clean benchmark model in which MRPK dispersion emerges even when firms are behaving efficiently.
- **Quantitative contribution.** Once the TFPR process and capital adjustment costs are taken seriously, the model explains a surprisingly large fraction of the observed variation in MRPK dispersion across industries and countries.
- **Interpretive contribution.** Static misallocation measures should not be read literally without first accounting for the dynamic benchmark.

7.2 Economic intuition

- Capital is slow-moving. When a firm receives a productivity shock, it cannot instantly reoptimize its capital stock.

- Therefore, even if the firm made the correct investment decision ex ante, it can look ex post as though capital is misallocated because its current MRPK differs from that of other firms.
- The more volatile the firm's environment, the more often this happens, and the larger the observed cross-sectional dispersion in MRPK becomes.
- The same logic is much weaker for labor and materials because those inputs are easier to adjust in the short run.

7.3 Takeaway

This paper is best read as a warning label for static misallocation exercises. Its message is not that distortions do not matter. Its message is that dispersion in MRPK is an equilibrium object in a dynamic environment, and volatility plus capital adjustment costs can generate a great deal of it even without policy wedges. For work on development, productivity, and resource allocation, this paper is the benchmark for what the "no-distortion" dynamic baseline should look like.